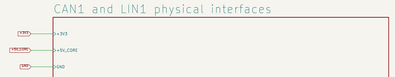
PowerBGDI electrical candidate / NOT RELEASED
Hierarchical KiCad project for the PowerBGDI stages, safety, motor candidates, sensor frontends and harness.
Outstanding channels and load qualification are tracked in REVISION_2026_03_21_ELECTRICAL.md.

MCU, power decoupling and debug



File: core_mcu.kicad_sch

CAN1 and LIN1 physical interfaces



File: core_comms.kicad_sch

Ignition permission and three dual gate drivers

File: ignition/ignition_common.kicad_sch

Ignition 1 – timed primary and OC latch

File: ignition/ignition_1.kicad_sch

Ignition 2 – timed primary and OC latch

File: ignition/ignition_2.kicad_sch

Ignition 3 – timed primary and OC latch

File: ignition/ignition_3.kicad_sch

Ignition 4 – timed primary and OC latch

File: ignition/ignition_4.kicad_sch

Ignition 5 – timed primary and OC latch

File: ignition/ignition_5.kicad_sch

Ignition 6 – timed primary and OC latch

File: ignition/ignition_6.kicad_sch

Buffered analog inputs 1..4

File: io/analog_1.kicad_sch

Buffered analog inputs 5..8

File: io/analog_2.kicad_sch

Buffered analog inputs 9..12

File: io/analog_3.kicad_sch

Buffered analog inputs 13..16

File: io/analog_4.kicad_sch

Buffered analog inputs 17..17

File: io/analog_5.kicad_sch

Hall and SENT receiver bank 1

File: io/digital_1.kicad_sch

Hall and SENT receiver bank 2

File: io/digital_2.kicad_sch

Differential VR crank receiver

File: io/crank_vr.kicad_sch

Direct protected low-side PI1, PI2, PI3, PI4

File: outputs/ls_bank_1.kicad_sch

Direct protected low-side PI5, PI6, VVT1, VVT2

File: outputs/ls_bank_2.kicad_sch

Direct protected low-side VVT3, VVT4, AUX_PWM1, AUX_PWM2

File: outputs/ls_bank_3.kicad_sch

Direct protected low-side AUX_PWM3, AUX_PWM4, AUX_PWM5, AUX_PWM6

File: outputs/ls_bank_4.kicad_sch

Direct protected low-side AUX_PWM7, AUX_PWM8, LS_AUX1, LS_AUX2

File: outputs/ls_bank_5.kicad_sch

Direct protected low-side LS_AUX3, LS_AUX4

File: outputs/ls_bank_6.kicad_sch

Low-side shared SPI and reset

File: outputs/ls_control.kicad_sch

Documented power sources across passive paths

File: power_sources.kicad_sch

GDI bank A reference candidate

File: gdi/stage_a.kicad_sch

GDI bank B reference candidate

File: gdi/stage_b.kicad_sch

Latched hardware permission

File: safety/safety_gate.kicad_sch

Four TE34 harness connectors

File: board/connectors.kicad_sch

CAN2 and LIN2 physical interfaces

File: interfaces/comms2.kicad_sch

Self-powered USB FS device: host VBUS never powers ECU

File: interfaces/usb_fs.kicad_sch

microSD SPI3 logging candidate

File: interfaces/sd_spi.kicad_sch

Dual sensor trackers and star returns

File: interfaces/sensor_power.kicad_sch

Additional Hall, flex and digital inputs bank 1

File: interfaces/digital_aux_1.kicad_sch

knock dual

File: interfaces_aux/knock_dual.kicad_sch

gdi/protection a

File: gdi/protection_a.kicad_sch

gdi/protection b

File: gdi/protection_b.kicad_sch

gdi/permission

File: gdi/permission.kicad_sch

wbo/wbo1 analog

File: wbo/wbo1_analog.kicad_sch

wbo/wbo1 control

File: wbo/wbo1_control.kicad_sch

wbo/wbo1 heater feed

File: wbo/wbo1_heater_feed.kicad_sch

logic power/logic 5v

File: logic_power/logic_5v.kicad_sch

logic power/logic 3v3

File: logic_power/logic_3v3.kicad_sch

motor4/dbw

File: motor4/dbw.kicad_sch

motor4/ewg

File: motor4/ewg.kicad_sch

motor4/spi

File: motor4/spi.kicad_sch

motor4/motor13 power

File: motor4/motor13_power.kicad_sch

motor4/motor24 power

File: motor4/motor24_power.kicad_sch

Each sheet uses explicit global net names. HARNESS names remain separate from MCU logic.
No ERC exclusions. Power flags have explicit independently checked source paths. DNP assembly flags must be preserved.
EDP enable, injector/pump models, mixing domains and full routing remain release blockers.

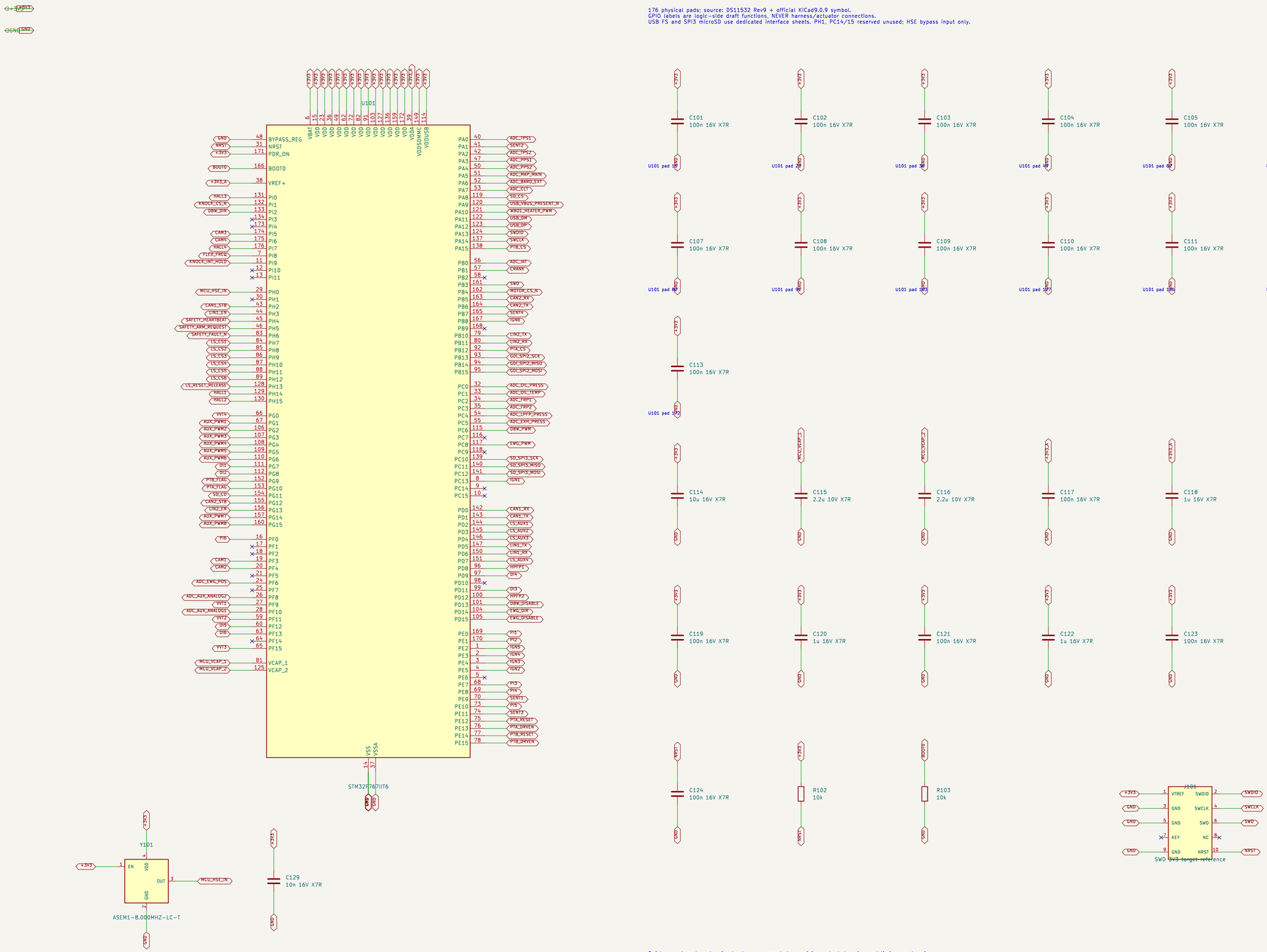
UNQUALIFIED: no manufacturing or engine approval

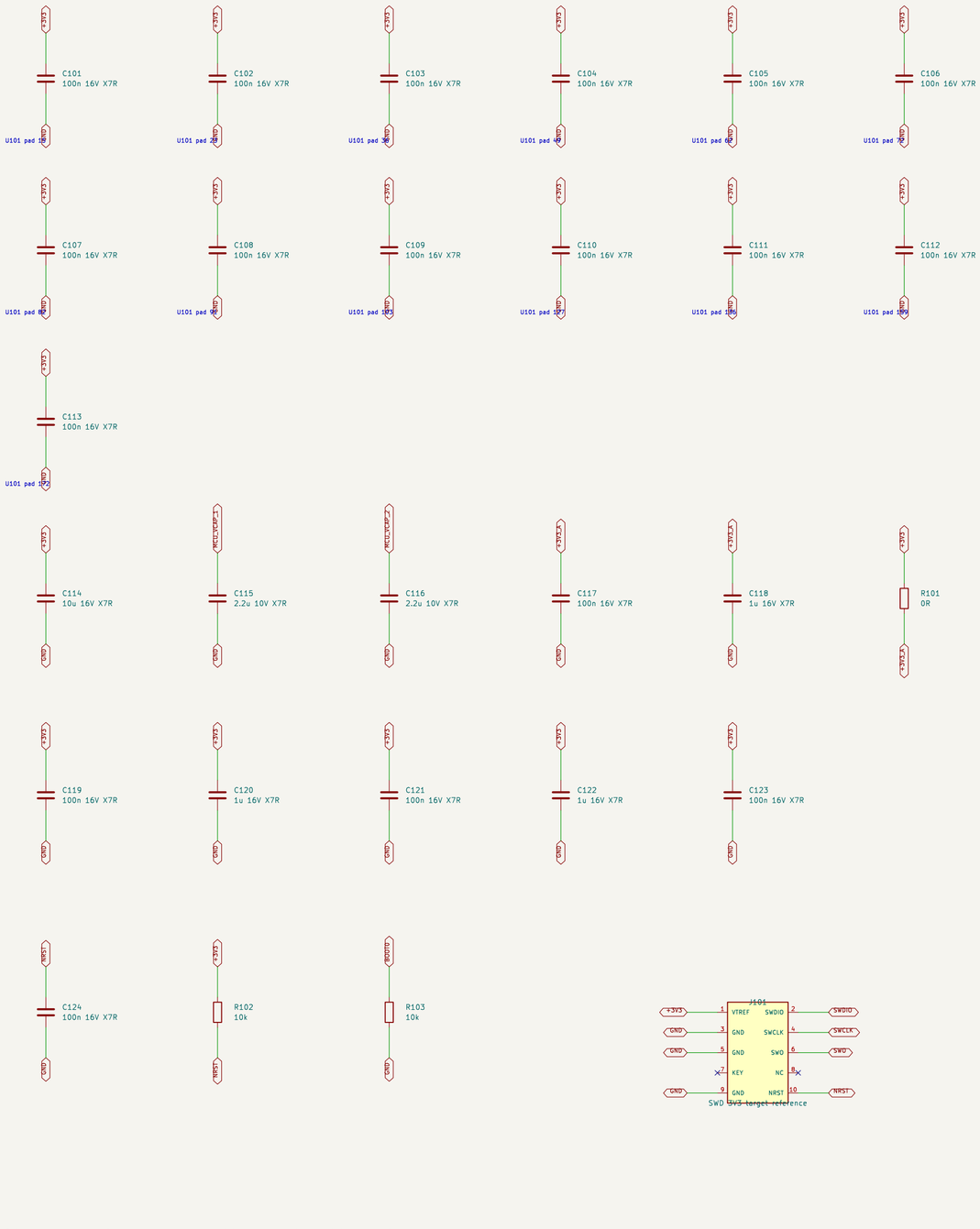
Sheet: /
File: PowerBGDI_Core.kicad_sch

Title: PowerBGDI core and interfaces

Size: A0	Number:	Date:	Rev: Engineering draft 0.1
KiCad 8.0.4 9.0.9			Id: 1/49

176 physical pads; source: DS11532 Rev9 + official KiCad9.0.9 symbol.
GPIO labels are logic-side draft functions, NEVER harness/actuator connections.
USB FS and SPI3 microSD use dedicated interface sheets. PH1, PC14/15 reserved unused; HSE bypass input only.





8MHz CMOS HSE: bypass mode required.
Oscillator load <=15pF, start<=3ms.
Clock tree/USB48MHz firmware review pending.

Safety, crank and analog frontends are connected; remaining output domains and timing need review.
IGN1=PC13 remains draft; low-drive restrictions require high-impedance buffer.
No MCU output is connected directly to a coil, injector, solenoid or vehicle wire.

UNQUALIFIED: no manufacturing or engine approval

Sheet: /MCU, power decoupling and debug/
File: core_mcu.kicad_sch

Title: MCU, power decoupling and debug

Size: A2 | Date: | **Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9 | Id: 2/49

+3V3

FROM_CORE

GND

R104
10k

R105
0R DNP

U103
TCAN1042HGVDRQ1

C127
100n 16V X7R

C128
100n 16V X7R

R106
120R DNP

D101
PESD2CAN

R107
4.7k

R108
10k

R109
0R DNP

U104
TLIN1029ADRQ1

R110
10k

R111
0R

C130
100n 50V X7R

C131
1u 50V X7R

D102
BAS21,215

R112
1k 1% 0.5W

C132
1n 100V COG

CAN1_H/L are vehicle-side nets. STB high by default; MCU PH2 pad43 enables low.
DNP termination only after network topology review. EMC/ESD bench qualification pending.
CAN2 and LIN2 use the comms2 sheet. PH2/PH3 GPIO additions are provisional schematic assignments.

LIN1 commander assembly: MCU PH3 pad44 enables high; default sleep.
One commander per bus. Responder assembly omits BOTH D102/R112.
VSUP4..36V operating,45V absolute. SMBJ24CA clamp38.9V is below absolute limit, not proof of operation during transients or pulse qualification.

UNQUALIFIED: no manufacturing or engine approval

Sheet: /CAN1 and LIN1 physical interfaces/
File: core_comms.kicad_sch

Title: CAN1 and LIN1 physical interfaces

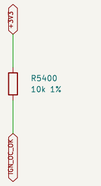
Size: A2 | Date: | **Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9 | Id: 3/49

PASSIVE PRIMARY OUTPUT ASSEMBLY

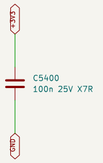
IGN_OC_OK is open drain, healthy high. Parent connects to HW_KILL_N.
No firmware-only current limit. Fault turn-off can cause an unintended spark.
Coil +12V feed must be removed on fault; see HW_IGNITION_B58.md.



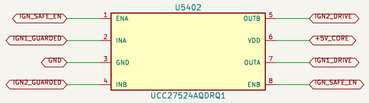


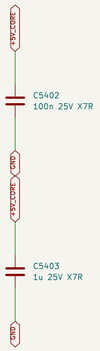








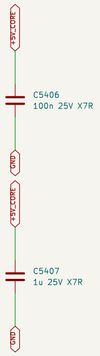












UNQUALIFIED: no manufacturing or engine approval

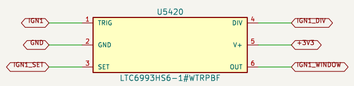
Sheet: /ignition permission and three dual gate drivers/
File: ignition_common.kicad_sch

Title: Ignition permission and three dual gate drivers

Size: A2	Date:	Rev: Engineering draft 0.1
KiCad E.D.A. 9.0.9		Id: 4/49

Channel 1 / passive assembly / unqualified candidate

LTC6993-1: rising edge, NON-retriggerable, 12.042 ms nominal window.
Gate = command AND window AND hardware permission. Not a dwell calibration.

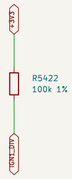


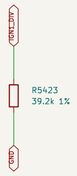


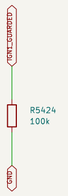




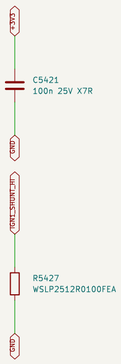


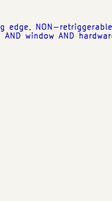


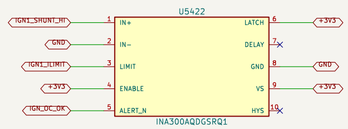
















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Overcurrent latches all eight channels off through IGN_OC_OK until power cycle.
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UNQUALIFIED: no manufacturing or engine approval

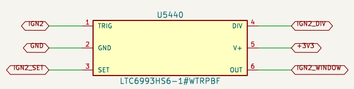
Sheet: /Ignition 1 – timed primary and OC latch/
File: ignition_1.kicad_sch

Title: Ignition 1 – timed primary and OC latch

Size: A2	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 5/49

Channel 2 / passive assembly / unqualified candidate

LTC6993-1: rising edge, NON-retriggerable, 12.042 ms nominal window.
Gate = command AND window AND hardware permission. Not a dwell calibration.



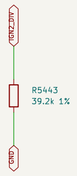


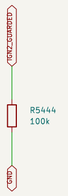


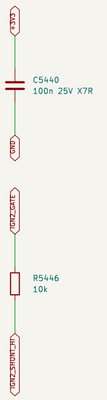


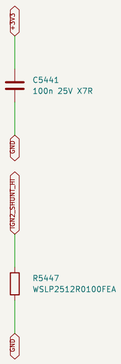


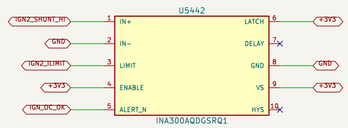
















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Hard cut-off may spark; soft shutdown and repetitive avalanche survival are NOT claimed.

UNQUALIFIED: no manufacturing or engine approval

Sheet: /Ignition 2 — timed primary and OC latch/
File: ignition_2.kicad_sch

Title: Ignition 2 — timed primary and OC latch

Size: A2	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 6/49

Channel 3 / passive assembly / unqualified candidate

LTC6993-1: rising edge, NON-retriggerable, 12.042 ms nominal window.
Gate = command AND window AND hardware permission. Not a dwell calibration.

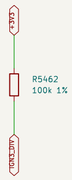


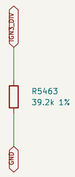


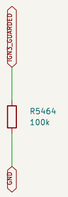


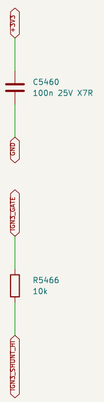






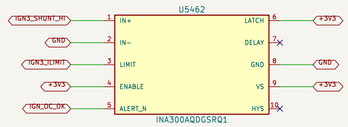
















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UNQUALIFIED: no manufacturing or engine approval

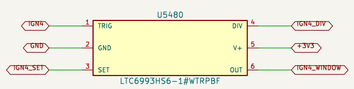
Sheet: /Ignition 3 — timed primary and OC latch/
File: ignition_3.kicad_sch

Title: Ignition 3 — timed primary and OC latch

Size: A2	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 7/49

Channel 4 / passive assembly / unqualified candidate

LTC6993-1: rising edge, NON-retriggerable, 12.042 ms nominal window.
Gate = command AND window AND hardware permission. Not a dwell calibration.

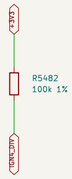




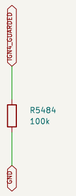








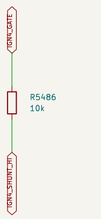


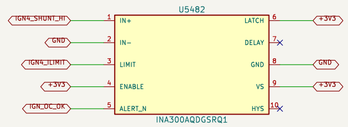
















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UNQUALIFIED: no manufacturing or engine approval

Sheet: /Ignition 4 — timed primary and OC latch/
File: ignition_4.kicad_sch

Title: Ignition 4 — timed primary and OC latch

Size: A2	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 8/49

Channel 5 / passive assembly / unqualified candidate

LTC6993-1: rising edge, NON-retriggerable, 12.042 ms nominal window.
Gate = command AND window AND hardware permission. Not a dwell calibration.



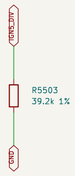


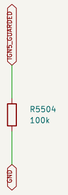




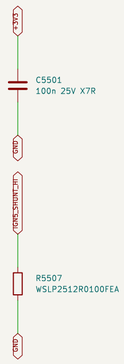




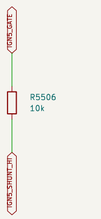


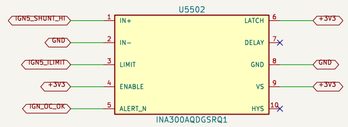
















IN+/IN- route as a Kelvin pair to shunt pads; do not carry ignition current in logic ground.
10us is typical comparator response, NOT a guaranteed maximum short-circuit delay.
Overcurrent latches all eight channels off through IGN_OC_OK until power cycle.
Hard cut-off may spark; soft shutdown and repetitive avalanche survival are NOT claimed.

UNQUALIFIED: no manufacturing or engine approval

Sheet: /Ignition 5 — timed primary and OC latch/
File: ignition_5.kicad_sch

Title: Ignition 5 — timed primary and OC latch

Size: A2	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 9/49

Channel 6 / passive assembly / unqualified candidate

LTC6993-1: rising edge, NON-retriggerable, 12.042 ms nominal window.
Gate = command AND window AND hardware permission. Not a dwell calibration.

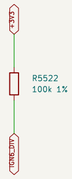


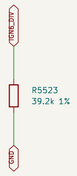


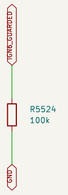


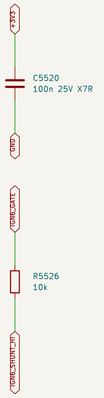


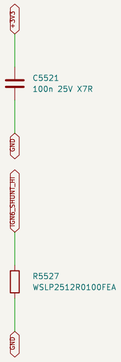




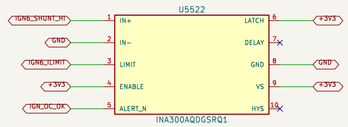


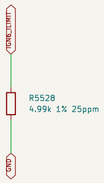














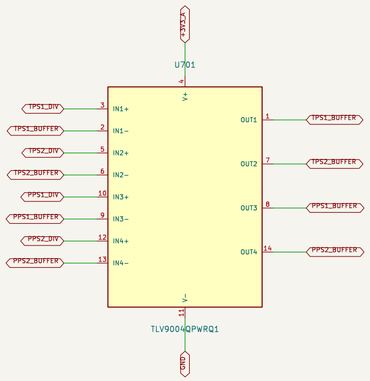
IN+/IN- route as a Kelvin pair to shunt pads; do not carry ignition current in logic ground.
10us is typical comparator response, NOT a guaranteed maximum short-circuit delay.
Overcurrent latches all eight channels off through IGN_OC_OK until power cycle.
Hard cut-off may spark; soft shutdown and repetitive avalanche survival are NOT claimed.

UNQUALIFIED: no manufacturing or engine approval

Sheet: /Ignition 6 — timed primary and OC latch/
File: ignition_6.kicad_sch

Title: Ignition 6 — timed primary and OC latch

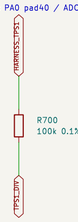
Size: A2	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 10/49





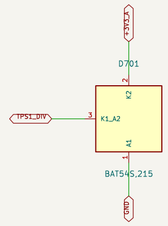


TPS1: PA0 pad40 / ADC1_IN0: normal input 0..5.5V





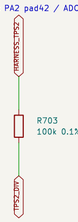


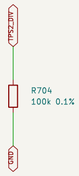




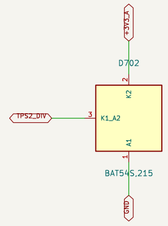


TPS2: PA2 pad42 / ADC1_IN2: normal input 0..5.5V





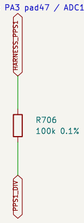


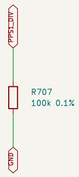




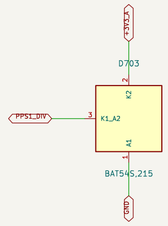


PPS1: PA3 pad47 / ADC1_IN3: normal input 0..5.5V

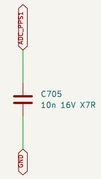




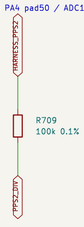


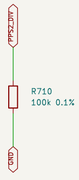




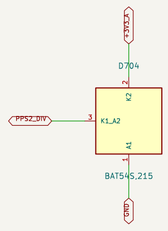


PPS2: PA4 pad50 / ADC1_IN4: normal input 0..5.5V













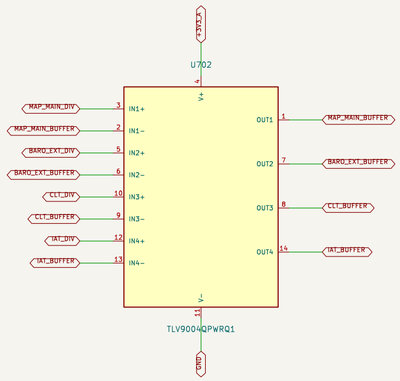
Nominal gain0.5; use2x voltage scaling in calibration.
RC, opamp offset, diode leakage, ADC sample time and reference error require analysis.
Fault clamps inject into3V3 rail; power-off/backfeed and simultaneous fault tests BLOCK RELEASE.
These are candidates for 0..5V sensors; no raw battery input or load may be connected.

UNQUALIFIED: no manufacturing or engine approval

Sheet: /Buffered analog inputs 1..4/
File: analog_1.kicad_sch

Title: Buffered analog inputs 1..4

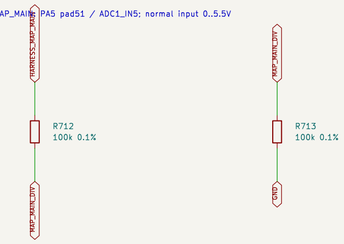
Size: A2 | Date: | **Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9 | Id: 11/49



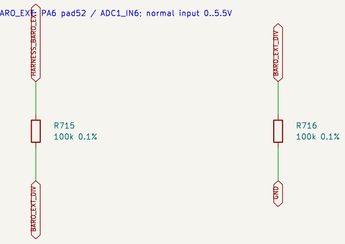




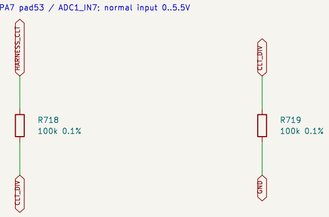
MAP_MAIN: PA5 pad51 / ADC1_IN5; normal input 0..5.5V



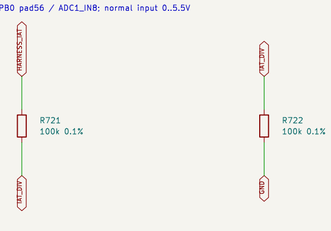
BARO_EXT: PA6 pad52 / ADC1_IN6; normal input 0..5.5V



CLT: PA7 pad53 / ADC1_IN7; normal input 0..5.5V



IAT: PB0 pad56 / ADC1_IN8; normal input 0..5.5V



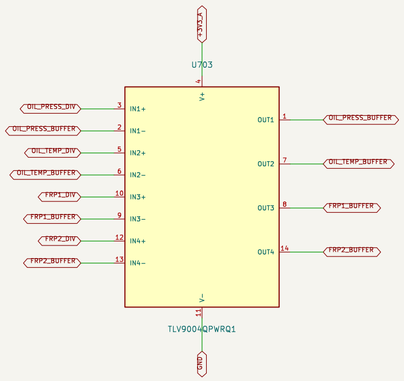
Nominal gain0.5; use2x voltage scaling in calibration.
RC, opamp offset, diode leakage, ADC sample time and reference error require analysis.
Fault clamps inject into3V3 rail; power-off/backfeed and simultaneous fault tests BLOCK RELEASE.
These are candidates for 0..5V sensors; no raw battery input or load may be connected.

UNQUALIFIED: no manufacturing or engine approval

Sheet: /Buffered analog inputs 5..8/
File: analog_2.kicad_sch

Title: Buffered analog inputs 5..8

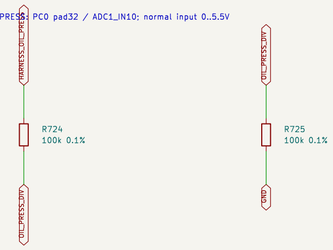
Size: A2	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 12/49

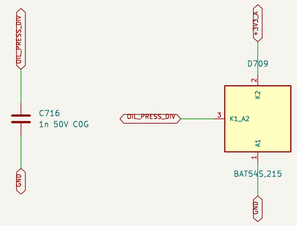


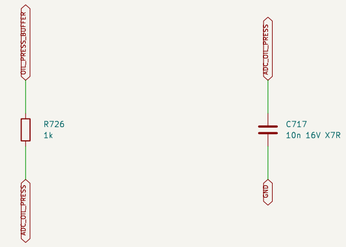




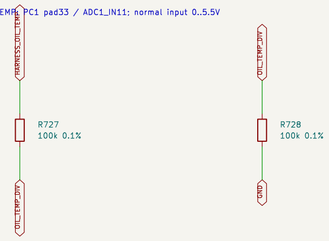
OIL_PRESS: PC0 pad32 / ADC1_IN10: normal input 0..5.5V

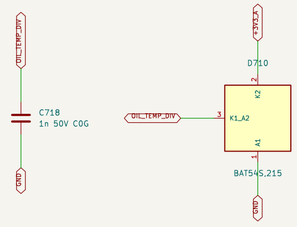


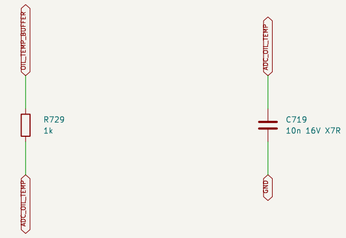




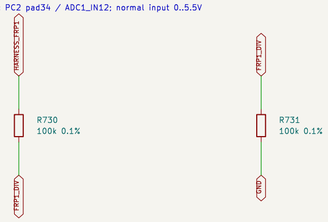
OIL_TEMP: PC1 pad33 / ADC1_IN11: normal input 0..5.5V

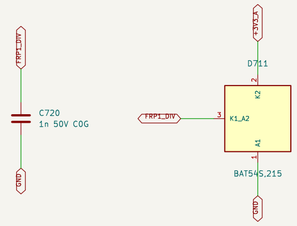


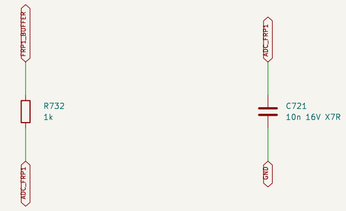




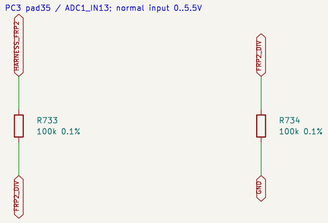
FRP1: PC2 pad34 / ADC1_IN12: normal input 0..5.5V

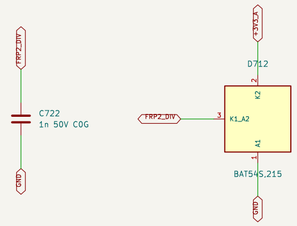


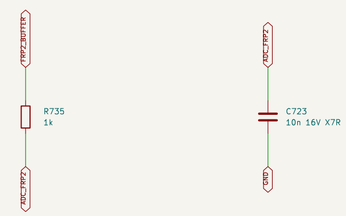




FRP2: PC3 pad35 / ADC1_IN13: normal input 0..5.5V







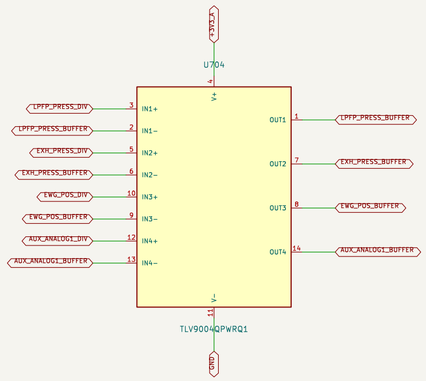
Nominal gain0.5; use2x voltage scaling in calibration.
RC, opamp offset, diode leakage, ADC sample time and reference error require analysis.
Fault clamps inject into3V3 rail; power-off/backfeed and simultaneous fault tests BLOCK RELEASE.
These are candidates for 0..5V sensors; no raw battery input or load may be connected.

UNQUALIFIED: no manufacturing or engine approval

Sheet: /Buffered analog inputs 9..12/
File: analog_3.kicad_sch

Title: Buffered analog inputs 9..12

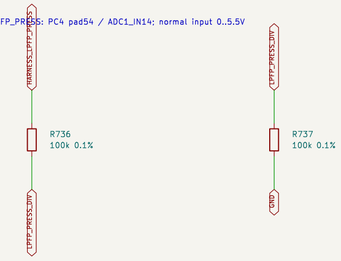
Size: A2 | Date: | **Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9 | Id: 13/49

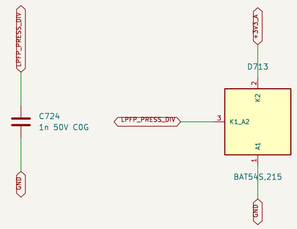


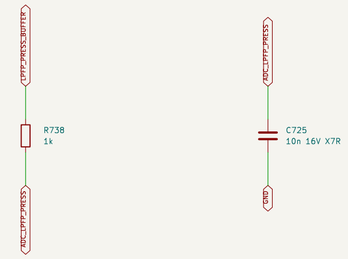




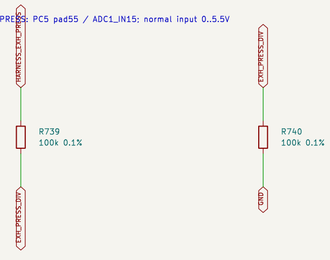
LPFP_PRESS: PC4 pad54 / ADC1_IN14; normal input 0..5.5V

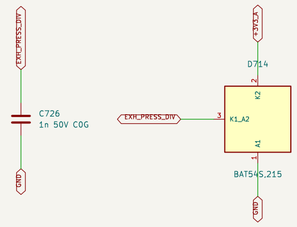


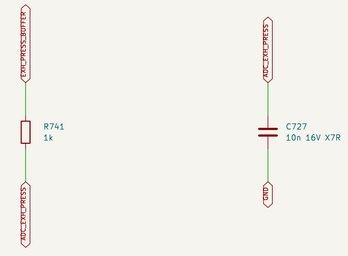




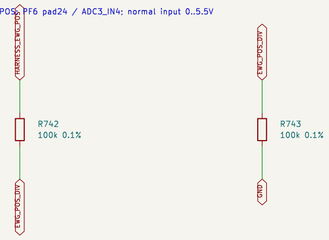
EXH_PRESS: PC5 pad55 / ADC1_IN15; normal input 0..5.5V

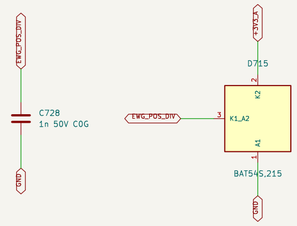


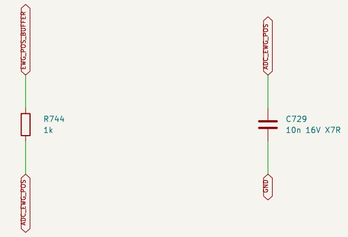




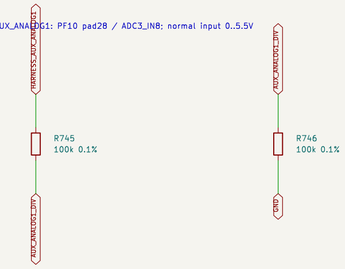
EWG_POS: PF6 pad24 / ADC3_IN4; normal input 0..5.5V

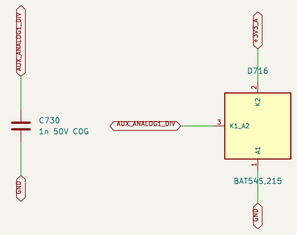


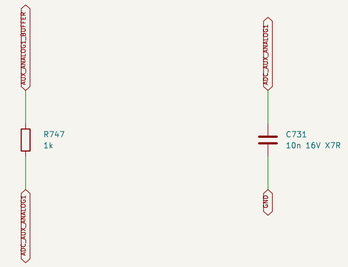




AUX_ANALOG1: PF10 pad28 / ADC3_IN8; normal input 0..5.5V







Nominal gain0.5; use2x voltage scaling in calibration.
RC, opamp offset, diode leakage, ADC sample time and reference error require analysis.
Fault clamps inject into3V3 rail; power-off/backfeed and simultaneous fault tests BLOCK RELEASE.
These are candidates for 0..5V sensors; no raw battery input or load may be connected.

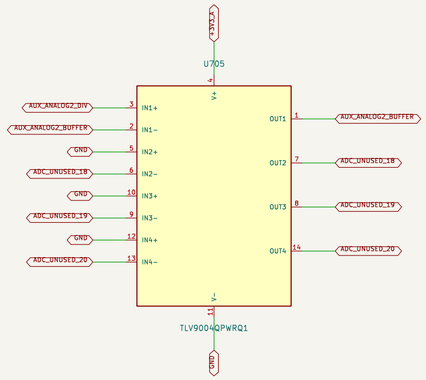
UNQUALIFIED: no manufacturing or engine approval

Sheet: /Buffered analog inputs 13..16/
File: analog_4.kicad_sch

Title: Buffered analog inputs 13..16

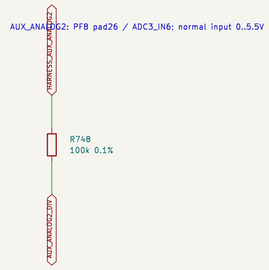
Size: A2	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 14/49

AUX_ANALOG2: PF8 pad26 / ADC3_IN6: normal input 0..5.5V

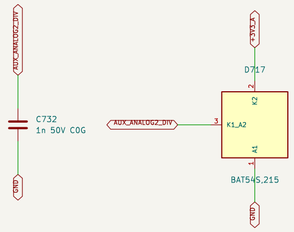




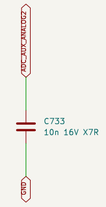


















Nominal gain0.5: use2x voltage scaling in calibration.
RC, opamp offset, diode leakage, ADC sample time and reference error require analysis.
Fault clamps inject into3V3 rail: power-off/backfeed and simultaneous fault tests BLOCK RELEASE.
These are candidates for 0..5V sensors; no raw battery input or load may be connected.

UNQUALIFIED: no manufacturing or engine approval

Sheet: /Buffered analog inputs 17..17/
File: analog_5.kicad_sch

Title: Buffered analog inputs 17..17

Size: A2 | Date: | **Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9 | Id: 15/49

U721

SN74LVC1HAQPWRQ1

C758
100n 16V X7R

CAM1: INVERTED at MCU; firmware edge/idle polarity required

R760
10k

R761
47k

C750
47p 100V C0G

D730
BAT545.215

R762
10k DNP

CAM2: INVERTED at MCU; firmware edge/idle polarity required

R763
10k

R764
47k

C751
47p 100V C0G

D731
BAT545.215

R765
10k DNP

CAM3: INVERTED at MCU; firmware edge/idle polarity required

R766
10k

R767
47k

C752
47p 100V C0G

D732
BAT545.215

R768
10k DNP

CAM4: INVERTED at MCU; firmware edge/idle polarity required

R769
10k

R770
47k

C753
47p 100V C0G

D733
BAT545.215

R771
10k DNP

SENT1: INVERTED at MCU; firmware edge/idle polarity required

R772
10k

R773
47k

C754
47p 100V C0G

D734
BAT545.215

R774
10k DNP

SENT2: INVERTED at MCU; firmware edge/idle polarity required

R775
10k

R776
47k

C755
47p 100V C0G

D735
BAT545.215

R777
10k DNP

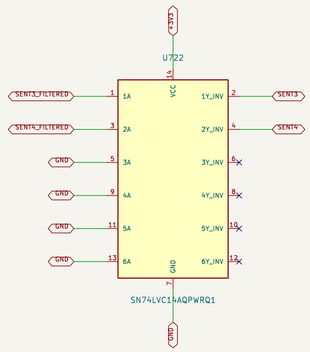
Hall/push-pull5V candidate only: CAM VR sensors require a different differential frontend.
Optional pullups DNP until sensor model confirmed. Every channel INVERTS.
SENT timing, input thresholds, clamp injection, cables and EMC remain unqualified.

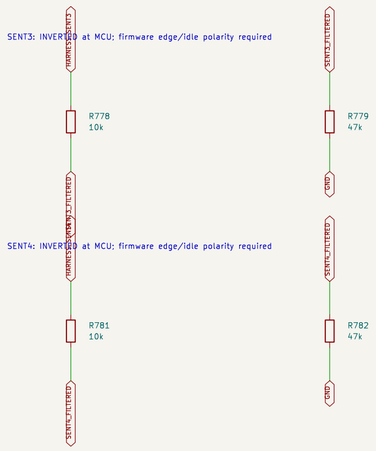
UNQUALIFIED: no manufacturing or engine approval

Sheet: /Hall and SENT receiver bank 1/
File: digital_1.kicad_sch

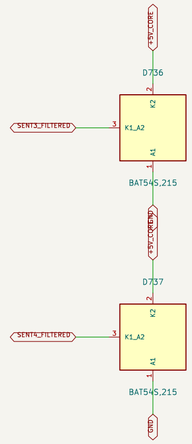
Title: Hall and SENT receiver bank 1

Size: A2 | Date: | **Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9 | Id: 16/49











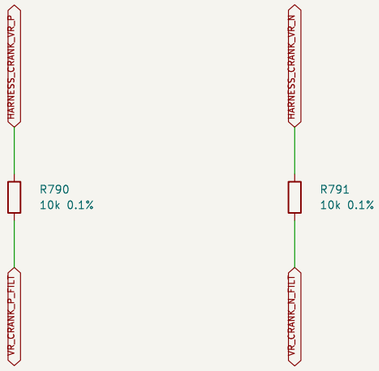
Hall/push-pull5V candidate only; CAM VR sensors require a different differential frontend.
Optional pullups DNP until sensor model confirmed. Every channel INVERTS.
SENT timing, input thresholds, clamp injection, cables and EMC remain unqualified.

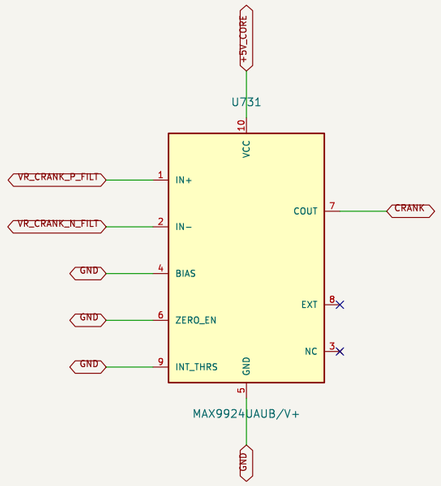
UNQUALIFIED: no manufacturing or engine approval

Sheet: /Hall and SENT receiver bank 2/
File: digital_2.kicad_sch

Title: Hall and SENT receiver bank 2

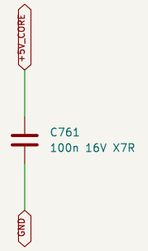
Size: A2	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 17/49

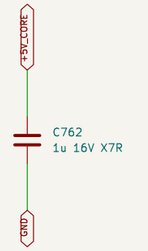












ModeA2: internal bias, adaptive threshold and zero−crossing enabled.
Supply4.5..5.5V; input absolute current +/−40mA is NOT an operating/protection target.
10k input resistors/filter are a reference−derived starting point; VR amplitude,
resistor pulse rating, cranking sensitivity, tooth polarity and EMI require bench review.
PB1 uses EXTI1 with scheduler timestamp: not a TIM3 capture channel shared with motors.

UNQUALIFIED: no manufacturing or engine approval

Sheet: /Differential VR crank receiver/
File: crank_vr.kicad_sch

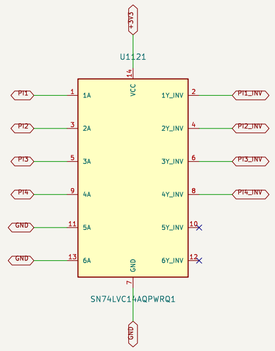
Title: Differential VR crank receiver

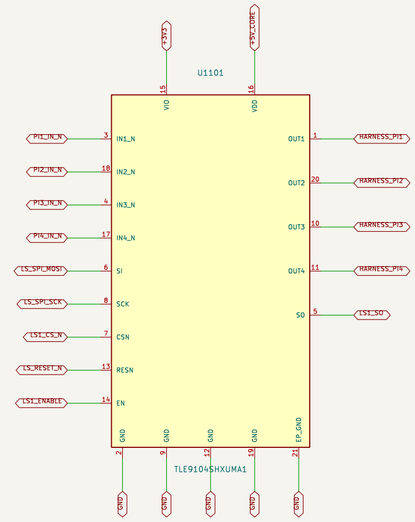
Size: A3	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 18/49







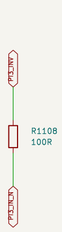


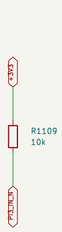




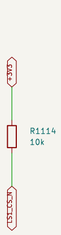


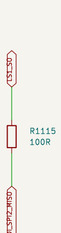


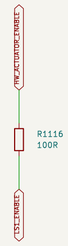




























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Candidate envelope for initial calculation: <=1.4A turn-off; repeated clamp energy <=7mJ, NOT an approved rating.
No low-impedance peak/hold injector support. VVT/AUX frequencies require scheduled GPIO implementation.
Driver fault thresholds, load L/R, simultaneous dissipation, harness fuses and board heat sinking remain unqualified.

UNQUALIFIED: no manufacturing or engine approval

Sheet: /Direct protected low-side PI1, PI2, PI3, PI4/
File: ls_bank_1.kicad_sch

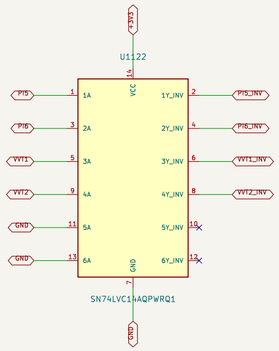
Title: Direct protected low-side PI1, PI2, PI3, PI4

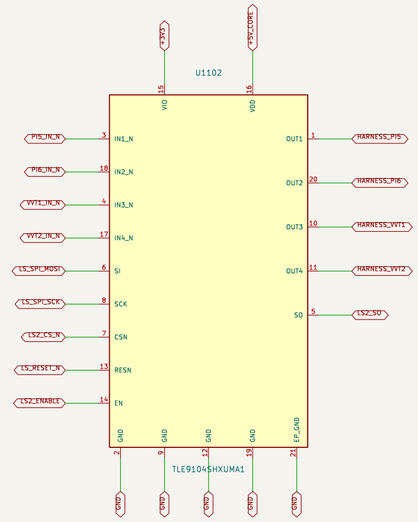
Size: A2	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 19/49

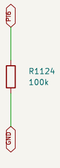




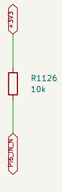


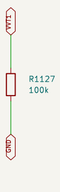












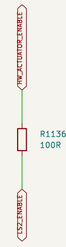


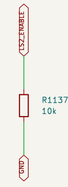






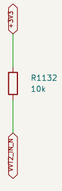




















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UNQUALIFIED: no manufacturing or engine approval

Sheet: /Direct protected low-side PI5, PI6, VVT1, VVT2/
File: ls_bank_2.kicad_sch

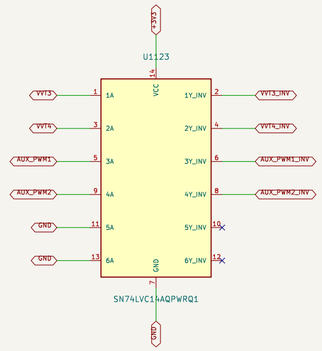
Title: Direct protected low-side PI5, PI6, VVT1, VVT2

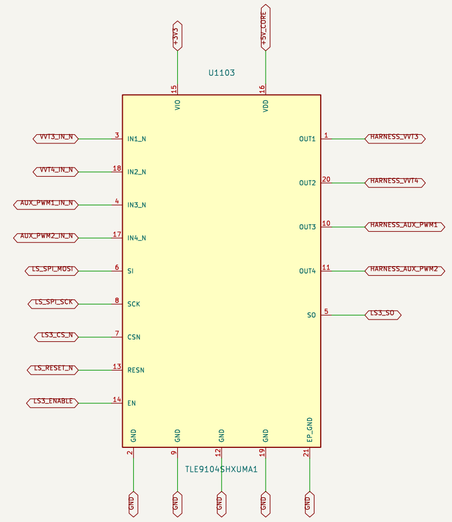
Size: A2	Date:	Rev: Engineering draft 0.1
KiCad E.D.A. 9.0.9		Id: 20/49





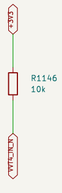


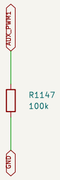












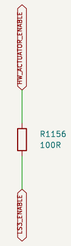




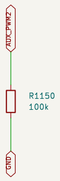




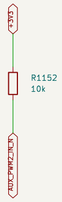




















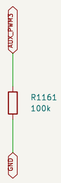
LOW-SIDE ONLY: external loads connect to separately fused +12V; return current through all driver GND/EP pads.
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UNQUALIFIED: no manufacturing or engine approval

Sheet: /Direct protected low-side VVT3, VVT4, AUX_PWM1, AUX_PWM2/
File: ls_bank_3.kicad_sch

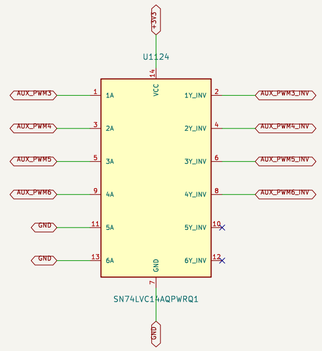
Title: Direct protected low-side VVT3, VVT4, AUX_PWM1, AUX_PWM2

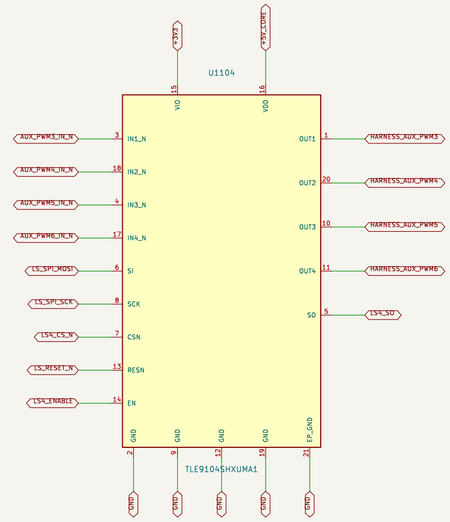
Size: A2 | Date: | **Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9 | Id: 21/49

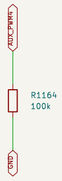






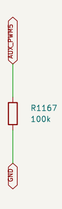








































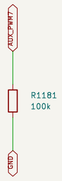
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Candidate envelope for initial calculation: <=1.4A turn-off; repeated clamp energy <=7mJ, NOT an approved rating.
No low-impedance peak/hold injector support. VVT/AUX frequencies require scheduled GPIO implementation.
Driver fault thresholds, load L/R, simultaneous dissipation, harness fuses and board heat sinking remain unqualified.

UNQUALIFIED: no manufacturing or engine approval

Sheet: /Direct protected low-side AUX_PWM3, AUX_PWM4, AUX_PWM5, AUX_PWM6/
File: ls_bank_4.kicad_sch

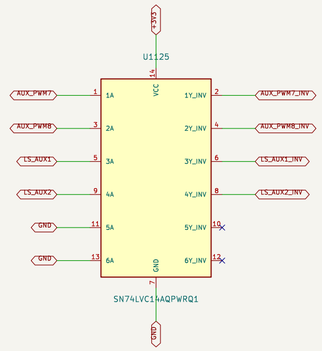
Title: Direct protected low-side AUX_PWM3, AUX_PWM4, AUX_PWM5, AUX_PWM

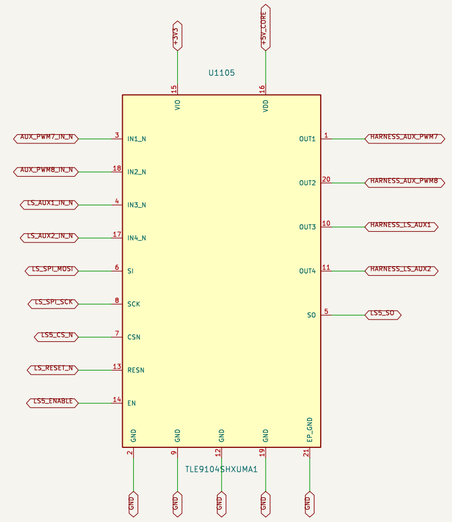
Size: A2	Date:	Rev: Engineering draft 0.1
KiCad E.D.A. 9.0.9		Id: 22/49

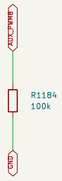




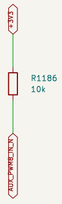


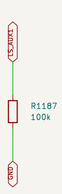








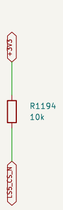






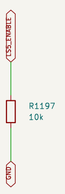






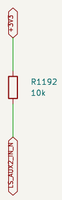




















LOW-SIDE ONLY: external loads connect to separately fused +12V; return current through all driver GND/EP pads.
3V3 MCU high -> inverter low -> load ON, only with hardware EN and SPI OUT_EN. MCU reset -> OFF.
Candidate envelope for initial calculation: <=1.4A turn-off; repeated clamp energy <=7mJ, NOT an approved rating.
No low-impedance peak/hold injector support. VVT/AUX frequencies require scheduled GPIO implementation.
Driver fault thresholds, load L/R, simultaneous dissipation, harness fuses and board heat sinking remain unqualified.

UNQUALIFIED: no manufacturing or engine approval

Sheet: /Direct protected low-side AUX_PWM7, AUX_PWM8, LS_AUX1, LS_AUX2/
File: ls_bank_5.kicad_sch

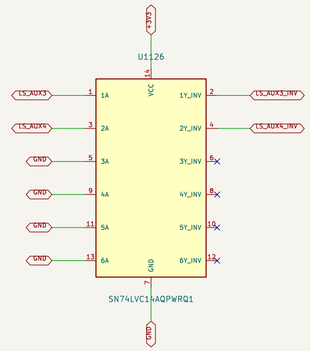
Title: Direct protected low-side AUX_PWM7, AUX_PWM8, LS_AUX1, LS_AUX2

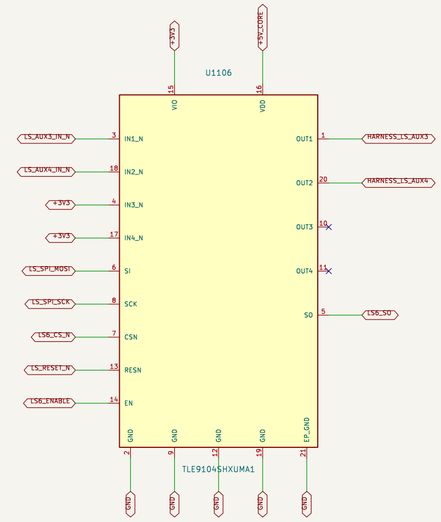
Size: A2	Date:	Rev: Engineering draft 0.1
KiCad E.D.A. 9.0.9		Id: 23/49





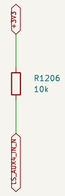




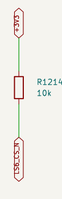


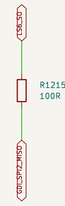


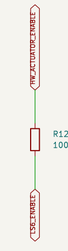






















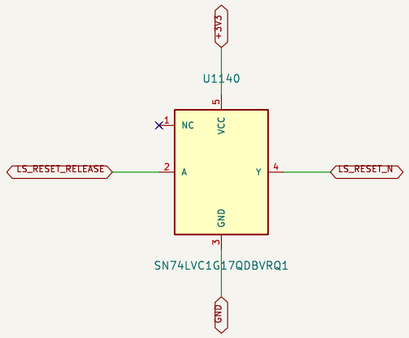
LOW-SIDE ONLY: external loads connect to separately fused +12V; return current through all driver GND/EP pads.
3V3 MCU high -> inverter low -> load ON, only with hardware EN and SPI OUT_EN. MCU reset -> OFF.
Candidate envelope for initial calculation: <=1.4A turn-off; repeated clamp energy <=7mJ, NOT an approved rating.
No low-impedance peak/hold injector support. VVT/AUX frequencies require scheduled GPIO implementation.
Driver fault thresholds, load L/R, simultaneous dissipation, harness fuses and board heat sinking remain unqualified.

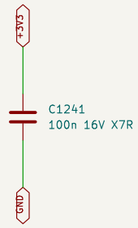
UNQUALIFIED: no manufacturing or engine approval

Sheet: /Direct protected low-side LS_AUX3, LS_AUX4/
File: ls_bank_6.kicad_sch

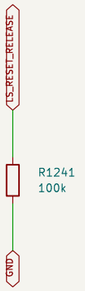
Title: Direct protected low-side LS_AUX3, LS_AUX4

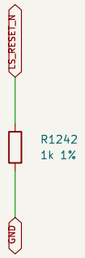
Size: A2	Date:	Rev: Engineering draft 0.1
KiCad E.D.A. 9.0.9		Id: 24/49













Startup: all22 commands low, sixCS high, reset-release low. Stabilize 3V3/5V; hold RESN low >=10us.
Release reset; verify POR, keep outputs disabled, set direct-drive/no-parallel CFG bits.
Set OUT_EN only after configuration readback and healthy watchdog; then arm hardware permission.
Maintain communication watchdog (default50ms); 16-bit odd-parity frames; bus mode/clock changed only with allCS high.
No automatic fault clearing/re-enable. External watchdog is not a per-injector maximum pulse timer.
SPI scheduling, rail sequencing, current/flyback/thermal and engine timing need hardware verification.

UNQUALIFIED: no manufacturing or engine approval

Sheet: /Low-side shared SPI and reset/
File: ls_control.kicad_sch

Title: Low-side shared SPI and reset

Size: A3	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 25/49

Power assertions document EXISTING physical source paths; no ERC exclusions.
Every path below is checked against the exported real circuit. Voltage/current/startup qualification remains separate.

PWR_FLAG
GND
External battery return at harness; not a self-powered synthetic rail

PWR_FLAG
+3V3
TPS62130A SW output through L5300

PWR_FLAG
VBAT_LOGIC_PROTECTED
Main externally fused input through series diode D5200

PWR_FLAG
LIN2_VSUP
Protected battery through fitted 0R R1506

PWR_FLAG
SENSOR_B_IN
Protected battery through D1531, anode2 to cathode1

PWR_FLAG
VBAT_GDI_A
Protected bank output through second MOSFET

PWR_FLAG
PTA_VCCIO
Main3V3 through fitted source sense resistor, power pads1/4

PWR_FLAG
VBAT_GDI_B
Protected bank output through second MOSFET

PWR_FLAG
PTB_VCCIO
Main3V3 through fitted source sense resistor, power pads1/4

PWR_FLAG
MOTOR13_VS
Motor protected output through second MOSFET; dependent on enable

PWR_FLAG
MOTOR24_VS
Motor protected output through second MOSFET; dependent on enable

PWR_FLAG
+5V_CORE
LM76003 SW output through L5200

PWR_FLAG
+3V3_A
Main 3V3 through fitted R101 filter feed

PWR_FLAG
LIN1_VSUP
Protected battery through fitted 0R R111

PWR_FLAG
SENSOR_A_IN
Protected battery through D1530, anode2 to cathode1

PWR_FLAG
GDI_A_MID
Protected-input MOSFET midpoint; gate/state/thermal qualification is separate

PWR_FLAG
PTA_VBOOST
Reference PT-controlled boost converter, rectifier cathode D12.2; not a guaranteed voltage

PWR_FLAG
GDI_B_MID
Protected-input MOSFET midpoint; gate/state/thermal qualification is separate

PWR_FLAG
PTB_VBOOST
Reference PT-controlled boost converter, rectifier cathode D12.2; not a guaranteed voltage

PWR_FLAG
MOTOR13_MID
Motor protected-input MOSFET midpoint; dependent on enable

PWR_FLAG
MOTOR24_MID
Motor protected-input MOSFET midpoint; dependent on enable

UNQUALIFIED: no manufacturing or engine approval

Sheet: /Documented power sources across passive paths/
File: power_sources.kicad_sch

Title: Documented power sources across passive paths

Size: A2 | Date: | **Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9 | Id: 26/49

R299 remains DNP pending PT electrical/shutdown qualification.
HPFP START6 is wired; loaded microcode/mask still requires verification.
Hardware permission and protected battery input are on separate GDI sheets.

U201
MC33PT2001AE

Independent 5V candidate: 7–20V regulation range; 35V ABS MAX is not a surge rating.
PT2001 VCC5/VCCIO current UNKNOWN; 500mA requires adequate heatsinking.
VBAT_GDI_A comes from LM74800 protection sheet; external fuse and transient tests required.

Hardware inhibit, microcode, load mapping and thermal design BLOCKED
Derived from rusEFI GDI4 rev.d; license unresolved; values not load-qualified

Sheet: /GDI bank A reference candidate/
File: stage_a.kicad_sch

Title: Power6GDI Stage A – 3 DI + 1 HPFP candidate

Size: A3 | Date: 2026-10-21 | Rev: engineering draft 0.1
KiCad E.D.A. 9.0.9 | Id: 27/49

R399 remains DNP pending PT electrical/shutdown qualification.
HPFP START6 is wired; loaded microcode/mask still requires verification.
Hardware permission and protected battery input are on separate GDI sheets.

U301
MC33PT2001AE

Independent 5V candidate: 7–20V regulation range; 35V ABS MAX is not a surge rating.
PT2001 VCC5/VCCIO current UNKNOWN; 500mA requires adequate heatsinking.
VBAT_GDI_B comes from LM74800 protection sheet; external fuse and transient tests required.

U349
NCV78M05ABDTRKG

Hardware inhibit, microcode, load mapping and thermal design BLOCKED
Derived from rusEFI GDI4 rev.d; license unresolved; values not load-qualified

Sheet: /GDI bank B reference candidate/	
File: stage_b.kicad_sch	
Title: Power6GDI Stage B – 3 DI + 1 HPFP candidate	
Size: A3	Date: 2026-10-21
KiCad E.D.A. 9.0.9	Id: 28/49
	Rev: engineering draft 0.1

ENGINEERING CANDIDATE — NOT QUALIFIED FOR ACTUATION

3.3 V logic only. Hardware faults asynchronously clear enable; recovery requires a fresh ARM rising edge.
Ignition OC joins HW_KILL_N through R6500. Per-load inhibition lives on driver sheets.
64.208 ms nominal watchdog is a backup CPU monitor, NOT a coil dwell/current limiter.



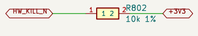








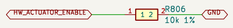


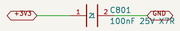




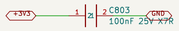


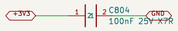




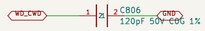


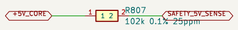


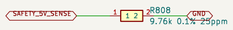


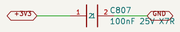














HW_KILL_N: local 3.3 V open-drain/contact-to-ground boundary only; no harness/12 V direct connection.
U806 monitors divided 5V independently of regulator PG. Fault release never substitutes for a fresh ARM edge.
C801..C804 at U801..U804; C807 at U806. Dynamic brownout/partial-power tests remain required.

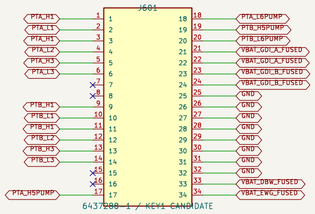
UNQUALIFIED: no manufacturing or engine approval

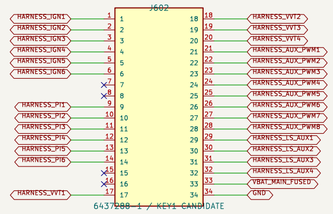
Sheet: /Latched hardware permission/
File: safety_gate.kicad_sch

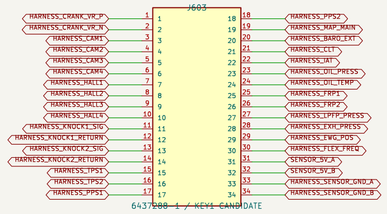
Title: Watchdog and latched global actuator inhibit

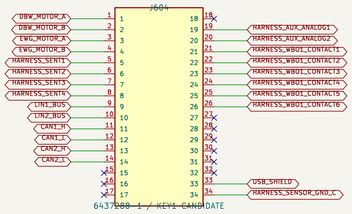
Size: A2	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 29/49

Four34-way SUPERSEAL1.0 headers. 4xKEY1 is provisional and NOT protected against cross-mating.
HARNESS_ prefix separates vehicle-side pins from MCU-side logic. Only connector_aliases.csv joins reviewed circuits.
No pin current/thermal qualification, high-side bank allocation, loom approval, EMC or engine approval.









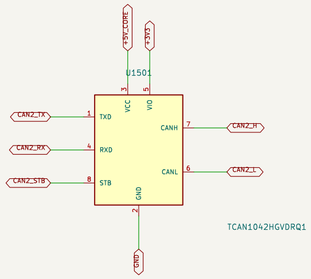
UNQUALIFIED: no manufacturing or engine approval

Sheet: /Four TE34 harness connectors/
File: connectors.kicad_sch

Title: Power6GDI TE34 harness boundary / UNQUALIFIED

Size: A2	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 30/49



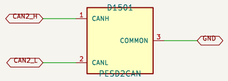




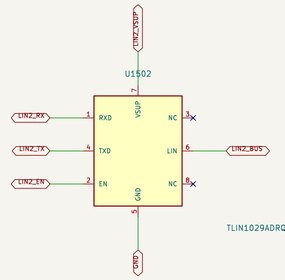








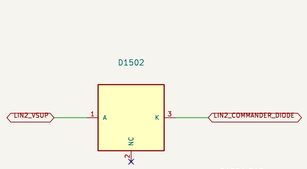






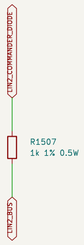












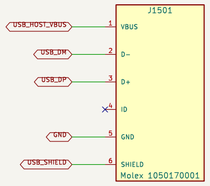
CAN2 default standby; LIN2 commander assembly, default sleep. MCU enables and schedules bus.
Only one commander pullup per LIN bus. Remove BOTH D1502/R1507 for responder-only assembly.
CAN2 uses PB5/PB6; SENT3 uses PE11 pad74. Shared timer, CAN filters and firmware remain pending.
No automotive pulse/ESD/EMC or fault qualification is implied by this candidate.

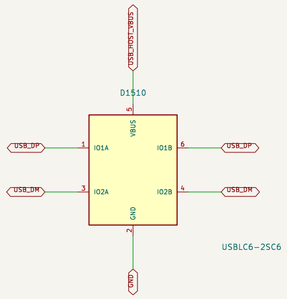
UNQUALIFIED: no manufacturing or engine approval

Sheet: /CAN2 and LIN2 physical interfaces/
File: comms2.kicad_sch

Title: CAN2 and LIN2 physical interfaces

Size: A2	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 31/49

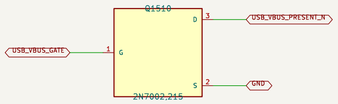


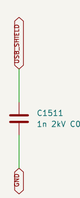












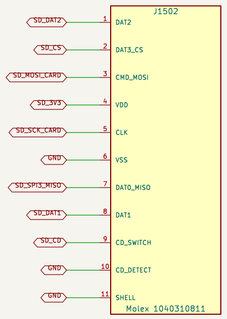
Firmware: device-only mode, PA11/PA12 AF10, PA9 GPIO low=VBUS present; native VBUS detect disabled.
No external D+ pullup or series termination is fitted: STM32F767 FS PHY includes the required functions.
Host VBUS feeds ESD reference and MOSFET gate ONLY; it must never be merged with +5V_CORE.
USB 48MHz clock, descriptors, attach/detach with ECU off, EMC and connector sealing remain unqualified.

UNQUALIFIED: no manufacturing or engine approval

Sheet: /Self-powered USB FS device: host VBUS never powers ECU/
File: usb_fs.kicad_sch

Title: Self-powered USB FS device: host VBUS never powers ECU

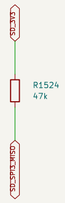
Size: A2	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 32/49



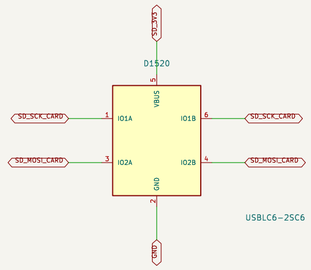


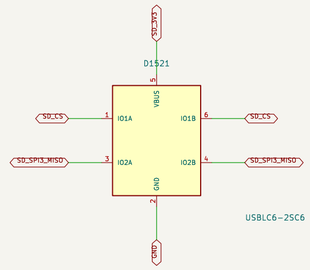












SPI mode only: PC10/11/12 AF6, PA8 chip select; PG11 card-detect polling, active low.
3V3 card supply and MCU rail are shared. Card peak/write current, inrush, LDO thermal budget and power-fail data integrity are unknown.
No UHS/1.8V signaling, SDMMC or hot-plug qualification. Internal service socket: Molex −25..85C rating limits the assembly.

UNQUALIFIED: no manufacturing or engine approval

Sheet: /microSD SPI3 logging candidate/
File: sd_spi.kicad_sch

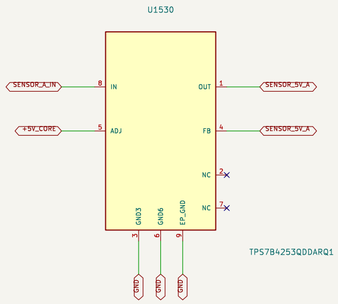
Title: *microSD SPI3 logging candidate*

Size: A2	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 33/49









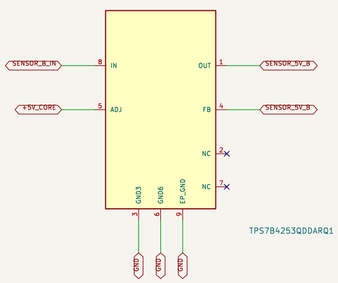
























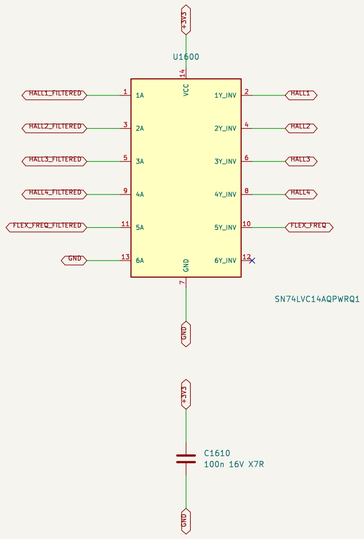
OUT=FB: unity tracking. ADJ reference+enable: no fault GPIO exists.
Input/output abs45V: upstream clamp and cable-inductance overshoot tests required.
At14V/5V.100mA dissipates about0.9W per tracker;300mA about2.7W. Neither is a qualified load budget.
Ground return links are NOT fuses. Star placement, ground offset, thermal, cold crank and fault tests block release.

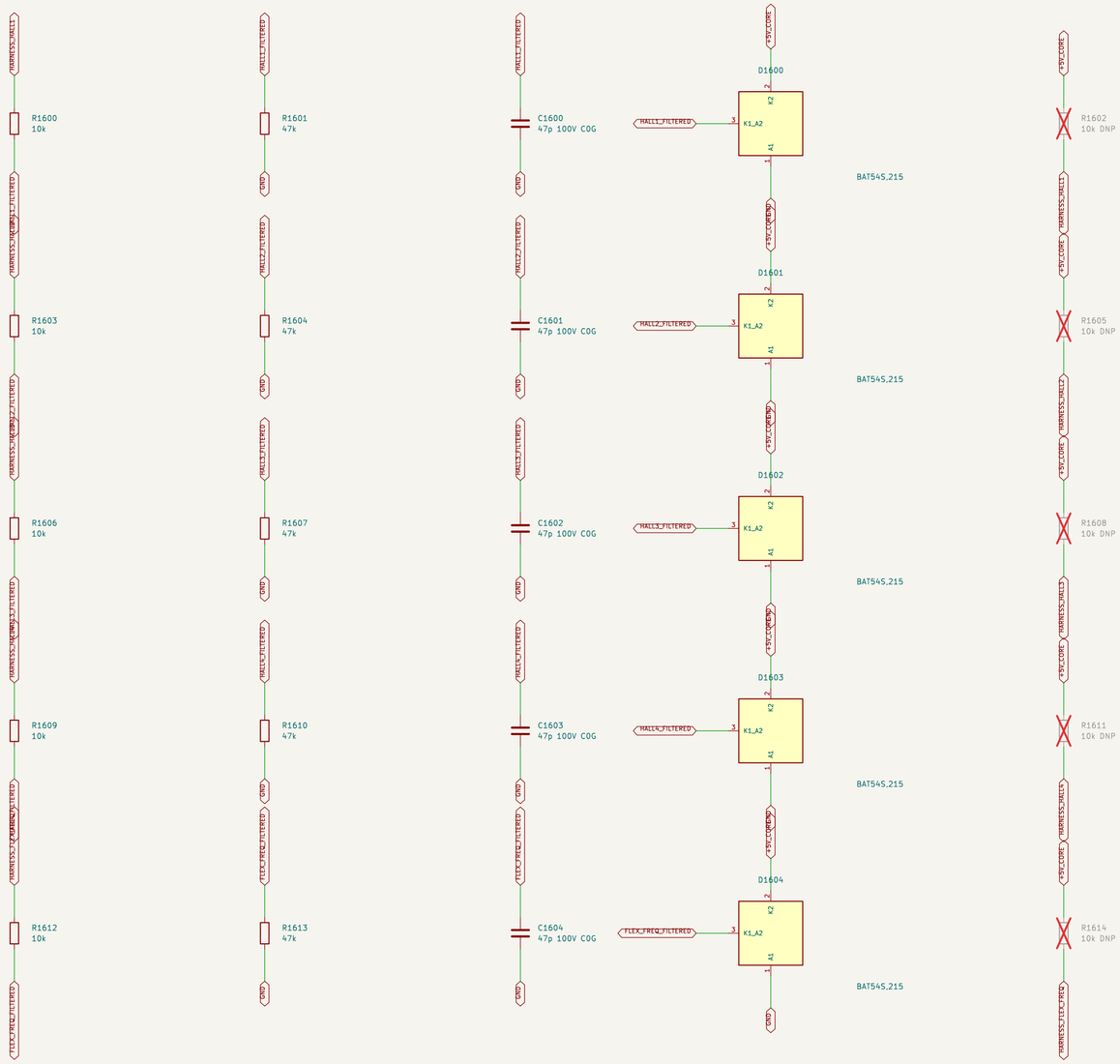
UNQUALIFIED: no manufacturing or engine approval

Sheet: /Dual sensor trackers and star returns/
File: sensor_power.kicad_sch

Title: Dual sensor trackers and star returns

Size: A2	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 34/49





HALL1..4 and FLEX_FREQ use EXTI14/15/0/7/8. Former DIN5..8 connector sites are NC in the six-cylinder target.
Input low/high thresholds, cable faults, power-off rail injection and flex frequency timing are not qualified.
Every active channel inverts. Optional open-drain pullups remain DNP until actual sensor models are known.

UNQUALIFIED: no manufacturing or engine approval

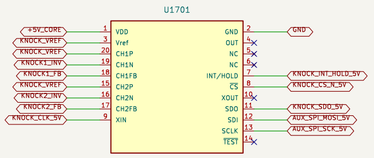
Sheet: /Additional Hall, flex and digital inputs bank 1/
File: digital_aux_1.kicad_sch

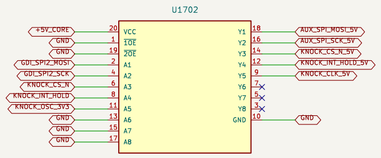
Title: Additional Hall, flex and digital inputs bank 1

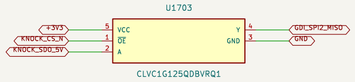
Size: A2	Date:	Rev: Engineering draft 0.1
KiCad E.D.A. 9.0.9		Id: 35/49

ENGINEERING CANDIDATE — NOT MANUFACTURING OR ENGINE APPROVED

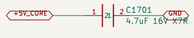
SPI digital readout: multiplexed dual sensors, not simultaneous acquisition. MCU ADC pins are not used.
Own 8 MHz oscillator; no added load on MCU oscillator. Sensor gain/window/filter/calibration remain unqualified.

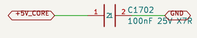


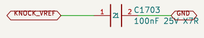


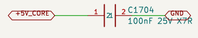


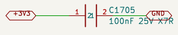


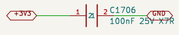






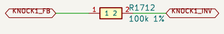


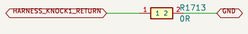






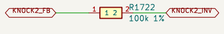


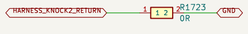


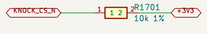
















UNQUALIFIED: no manufacturing or engine approval

Sheet: /knock dual/
File: knock_dual.kicad_sch

Title: Two-input TPIC8101 and 3.3/5 V serial interface

Size: A2	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 36/49

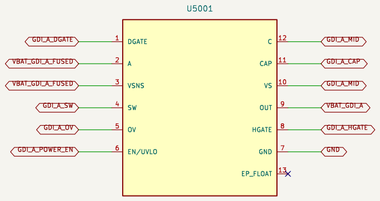
GDI A INPUT: separate externally fused battery feed

Common-drain MOSFETs: input is Q1 SOURCE, midpoint joins DRAINS, output is Q2 SOURCE.

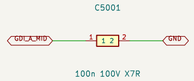
LM74800 exposed pad MUST FLOAT. Keep copper under pad isolated; do not use GND thermal vias.

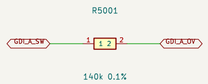


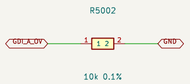


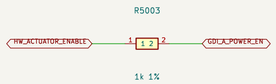


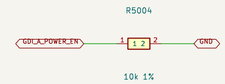


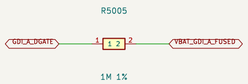






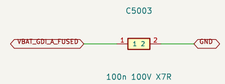


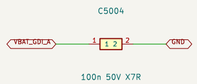




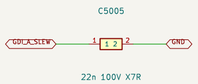


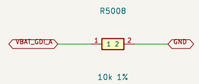












OV disconnect protects the 5V regulator normal input envelope; dynamic overshoot has NOT been measured.

External fuse is REQUIRED; this controller has no precision overcurrent limit. Fuse rating depends on load and harness.

Power removal does not instantly remove stored boost energy. DRVEN hardware inhibit remains a separate path.

Startup: hold DI/HPFP commands and DRVEN low; ARM bank power; wait for rails; reset/program/verify PT; then permit DRVEN.

UNQUALIFIED: no manufacturing or engine approval

Sheet: /gdi/protection a/
File: protection_a.kicad_sch

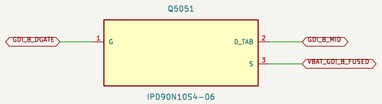
Title: GDI bank A: protected battery feed

Size: A2	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 37/49

GDI B INPUT: separate externally fused battery feed

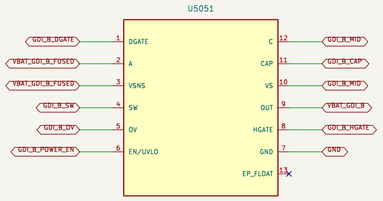
Common-drain MOSFETs: input is Q1 SOURCE, midpoint joins DRAINS, output is Q2 SOURCE.

LM74800 exposed pad MUST FLOAT. Keep copper under pad isolated; do not use GND thermal vias.

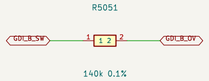


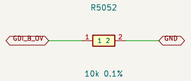


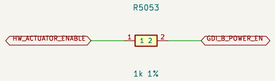


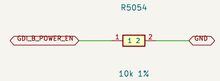


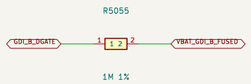






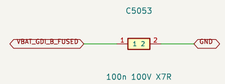






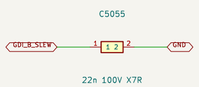


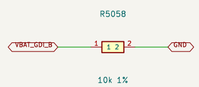












OV disconnect protects the 5V regulator normal input envelope; dynamic overshoot has NOT been measured.

External fuse is REQUIRED; this controller has no precision overcurrent limit. Fuse rating depends on load and harness.

Power removal does not instantly remove stored boost energy. DRVEN hardware inhibit remains a separate path.

Startup: hold DI/HPFP commands and DRVEN low; ARM bank power; wait for rails; reset/program/verify PT; then permit DRVEN.

UNQUALIFIED: no manufacturing or engine approval

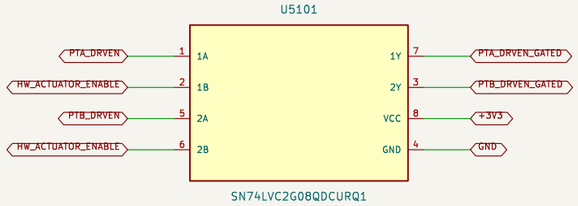
Sheet: /gdi/protection b/
File: protection_b.kicad_sch

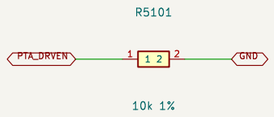
Title: GDI bank B: protected battery feed

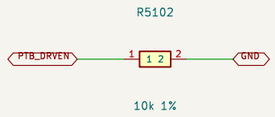
Size: A2	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 38/49

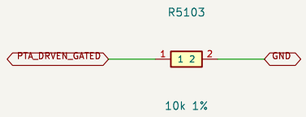
HARDWARE PERMISSION: stage request AND latched actuator enable

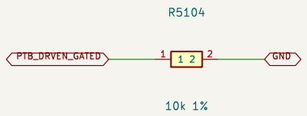
No extra MCU GPIOs. R299/R399 isolation links retain their explicit qualification hold.

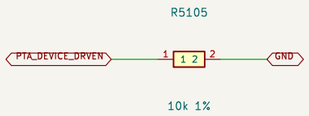


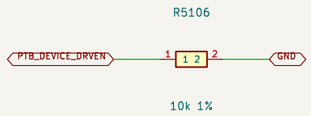


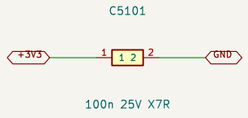












Gate never interprets FLAG0 as a fault without its programmed meaning. Per-bank CS, RESET and DRVEN stay distinct.

This sheet implements logic gating; PT electrical thresholds, power sequencing and pump shutdown configuration still need validation.

UNQUALIFIED: no manufacturing or engine approval

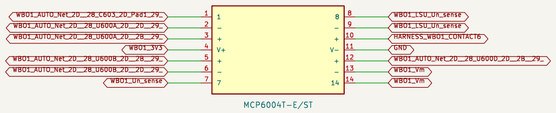
Sheet: /gdi/permission/
File: permission.kicad_sch

Title: Independent GDI permissions and default-off outputs

Size: A3	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 39/49

Hellen WBO 0.6 derived candidate; no engine / thermal / EMC approval

Pinned source f8be5db671b7a2739d0c5c2ee7c9a80c3022aa76. Design changes and remaining checks in README.





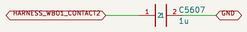




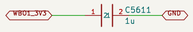


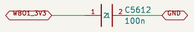








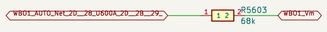






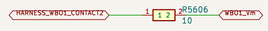


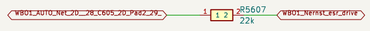








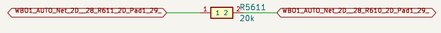


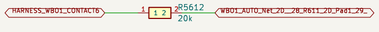


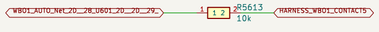


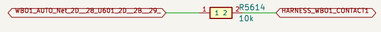


























UNQUALIFIED: no manufacturing or engine approval

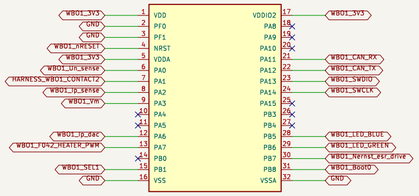
Sheet: /wbo/wbo1 analog/
File: wbo1_analog.kicad_sch

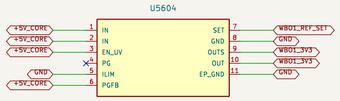
Title: WBO1: LSU4.9 pump and Nernst analog loop

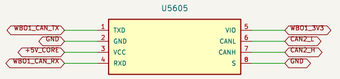
Size: A2	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 40/49

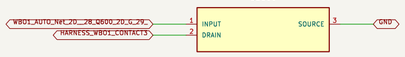
Hellen WBO 0.6 derived candidate; no engine / thermal / EMC approval

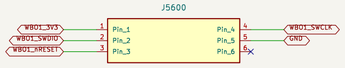
Pinned source f8be5db671b7a2739d0c5c2ee7c9a80c3022aa76. Design changes and remaining checks in README.





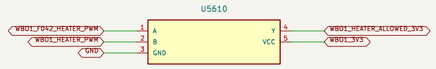




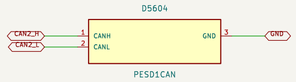












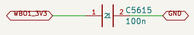








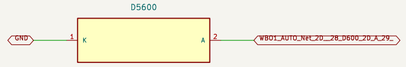


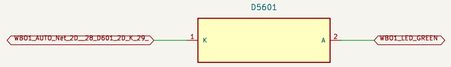


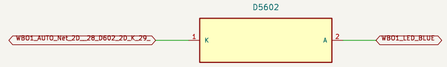


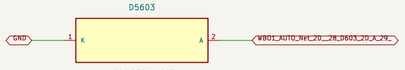








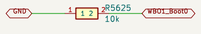














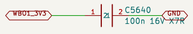


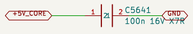






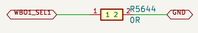






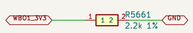


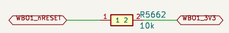


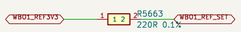












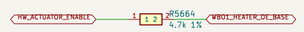














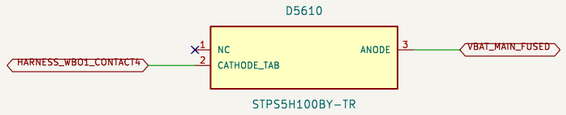
UNQUALIFIED: no manufacturing or engine approval

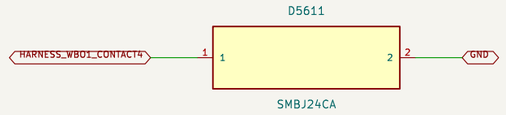
Sheet: /wbo/wbo1 control/
File: wbo1_control.kicad_sch

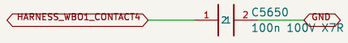
Title: WBO1: F042, CAN and hardware-inhibited heater

Size: A2	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 41/49

Heater current uses VBAT_MAIN_FUSED and dedicated power return; never +5V_CORE or VBAT_LOGIC_PROTECTED.









Normal sensor supply10.8..16.5V after diode. Control receives measured voltage over CAN.
Diode drop must be compensated in reported heater supply; current implementation has no local supply ADC.
Bosch LSU4.9 steady heater power7.5W/channel; cold start, fuse coordination and thermal tests pending.

UNQUALIFIED: no manufacturing or engine approval

Sheet: /wbo/wbo1 heater feed/
File: wbo1_heater_feed.kicad_sch

Title: WBO1: reverse protected heater supply

Size: A3	Date:	Rev: Engineering draft 0.1
KiCad E.D.A. 9.0.9		Id: 42/49

LOGIC 5V: replaces the complete old power_input sheet. Keep externally fused main battery input.

Protected battery also feeds two sensor trackers and LIN VSUP. Their current bypasses the 5V buck.

D5200



STPS5H100BY-TR

D5201



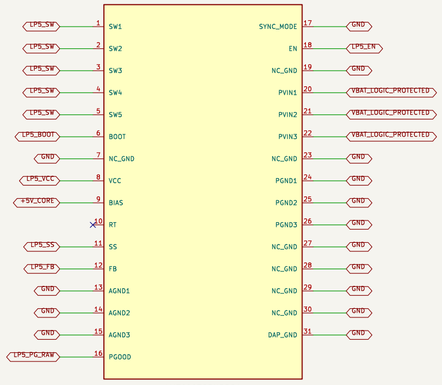
SMBJ24CA

L5200



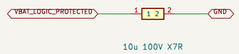
XAL7070-682MEC 6.8uH

U5200



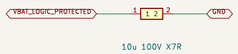
LM76003QRNPRQ1

C5200



10u 100V X7R

C5201



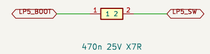
10u 100V X7R

C5202



47n 100V X7R

C5203



470n 25V X7R

C5204



2.2u 16V X7R

C5205



1u 16V X7R

C5206



22n 16V X7R

R5200



100k 0.1%

R5201



25k 0.1%

C5207



180p 50V C0G

R5202



470k 0.1%

R5203



100k 0.1%

R5204



100k 1%

R5205



1M 1%

C5208



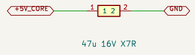
47u 16V X7R

C5209



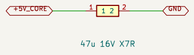
47u 16V X7R

C5210



47u 16V X7R

C5211



47u 16V X7R

Input range is conditional: nominal EN rise6.86V atprotected rail; cold crank below this turns rails off.

No ISO7637/load dump qualification; no independent overvoltage disconnect on sensor/LIN battery rail.

UNQUALIFIED: no manufacturing or engine approval

Sheet: /logic power/logic 5v/
File: logic_5v.kicad_sch

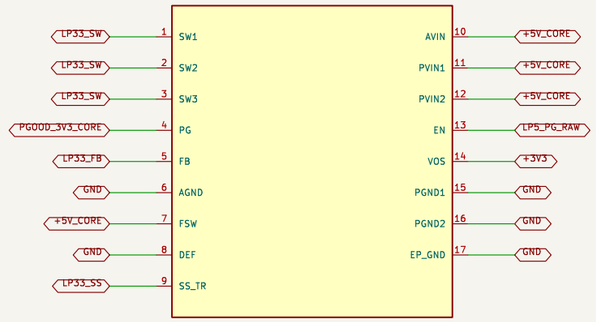
Title: Protected logic input and 5V / 2A design envelope

Size: A2	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 43/49

3V3: REPLACE old core_power/TLV1117 sheet; never populate both sources in parallel.

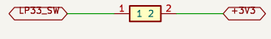
5V PGOOD directly enables TPS62130A. 3V3 PG is optional diagnostic; existing supervisor remains authoritative.

U5300



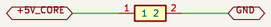
TPS62130AQRGTRQ1

L5300



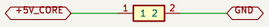
XAL5030-222MEC 2.2uH

C5300



10u 25V X7R

C5301



100n 25V X7R

C5302



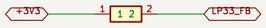
22u 16V X7R

C5303



3.3n 25V C0G

R5300



100k 0.1%

R5301



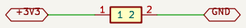
32k 0.1%

R5302



100k 1%

R5303



10k 1%

Target1A is a design allocation, not thermal qualification. 3V3 short removes5V if buck protection propagates.

UNQUALIFIED: no manufacturing or engine approval

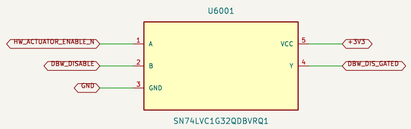
Sheet: /logic power/logic 3v3/
File: logic_3v3.kicad_sch

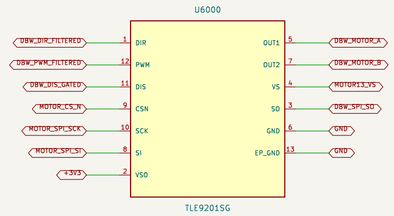
Title: Sequenced 3V3 / 1A design envelope

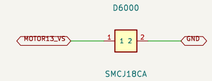
Size: A3	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 44/49

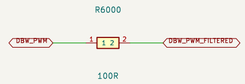
DBW: direct timer PWM / DIR / default-off DIS; 2A continuous DESIGN TARGET ONLY

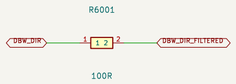
VS from dedicated protected motor feed; VSO=+3V3 logic. PWM=0 does not disconnect both motor terminals.

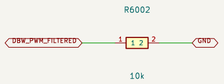


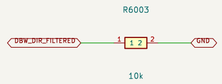


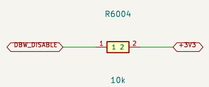


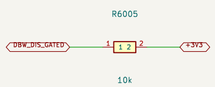












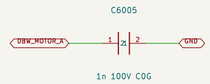


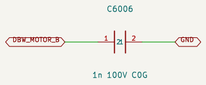












Place VS bypass and TVS close to pins4/6; EP13 solid GND with thermal vias. Power returns avoid sensor ground necks.

Current chopping6..10A is NOT a precise operating current limit; no current ADC. Stall timeout and position plausibility required.

Both PWM pins share TIM3 frequency. Use <=20kHz; begin bench bring-up at2kHz and validate final switching/EMI losses.

UNQUALIFIED: no manufacturing or engine approval

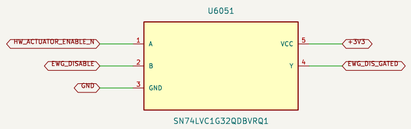
Sheet: /motor4/dbw/
File: dbw.kicad_sch

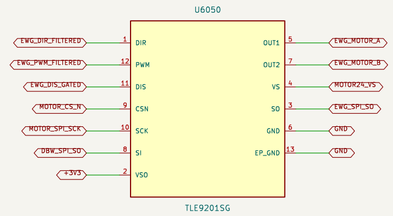
Title: DBW: TLE9201 motor bridge

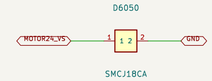
Size: A2	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 45/49

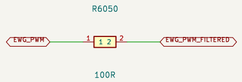
EWG: direct timer PWM / DIR / default-off DIS; 2A continuous DESIGN TARGET ONLY

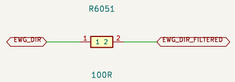
VS from dedicated protected motor feed; VSO=+3V3 logic. PWM=0 does not disconnect both motor terminals.

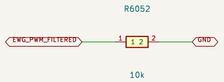


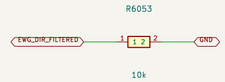


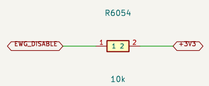


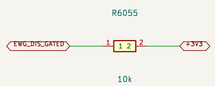










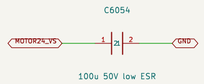


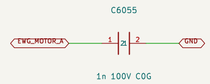














Place VS bypass and TVS close to pins4/6; EP13 solid GND with thermal vias. Power returns avoid sensor ground necks.

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Both PWM pins share TIM3 frequency. Use <=20kHz; begin bench bring-up at2kHz and validate final switching/EMI losses.

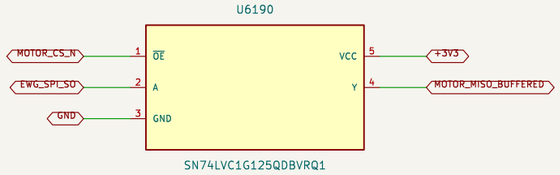
UNQUALIFIED: no manufacturing or engine approval

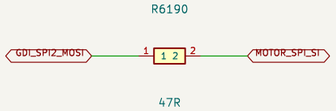
Sheet: /motor4/ewg/
File: ewg.kicad_sch

Title: EWG: TLE9201 motor bridge

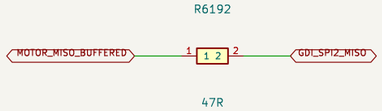
Size: A2	Date:	**Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9		Id: 46/49

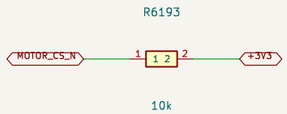
Shared SPI2: MOSI -> DBW -> EWG -> gated MISO

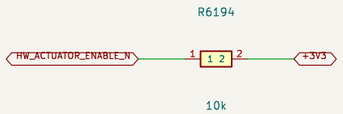


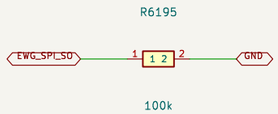














Mode1, MSB-first, <=1MHz initial configuration, 16 clocks per CS pulse; hold CS high >=2us between frames.

Send bytes EWG,DBW. Receive previous-frame replies in the same order. SIN bit must remain0.

Serialize with PT2001 transactions. Keep all other CS high; change SPI mode/rate only with every CS high.

UNQUALIFIED: no manufacturing or engine approval

Sheet: /motor4/spi/
File: spi.kicad_sch

Title: Motor daisy chain and shared SPI isolation

Size: A3	Date:	Rev: Engineering draft 0.1
KiCad E.D.A. 9.0.9		Id: 47/49

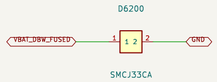
MOTOR13: one motor channel per externally fused connector feed; 2A continuous DESIGN TARGET

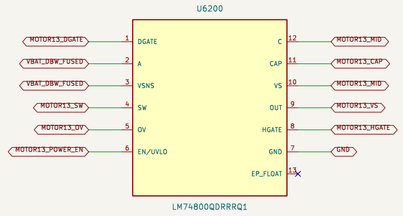
Input OV disconnect nominal18.465V; static maximum19.075V. No precision current limit in this supply stage.

LM74800 EP13 MUST FLOAT: no GND plane connection / no thermal vias. Input FET source faces battery.

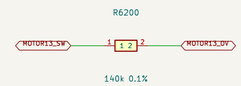


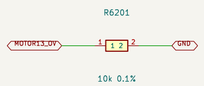




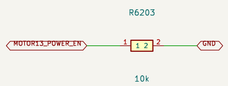




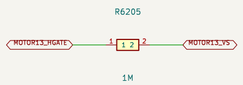




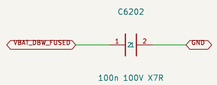








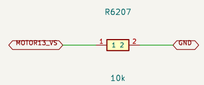












Ideal-diode input blocks regeneration to battery: 2x100uF plus local18V TVS absorb transient energy; load energy still UNKNOWN.

Global hardware latch removes supply AND forces DIS high; default-off EN10k. Residual bulk voltage is not immediate safe discharge.

No 24V jump-start operating claim. Verify OV turnoff overshoot, loss-of-battery braking, hot short and external fuse/contact budget.

UNQUALIFIED: no manufacturing or engine approval

Sheet: /motor4/motor13_power/
File: motor13_power.kicad_sch

Title: MOTOR13: independent protected feed

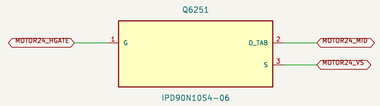
Size: A2 | Date: | **Rev: Engineering draft 0.1**
KiCad E.D.A. 9.0.9 | Id: 48/49

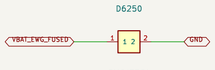
MOTOR24: one motor channel per externally fused connector feed; 2A continuous DESIGN TARGET

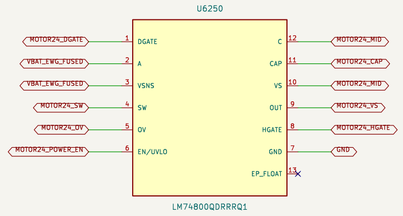
Input OV disconnect nominal18.465V; static maximum19.075V. No precision current limit in this supply stage.

LM74800 EP13 MUST FLOAT: no GND plane connection / no thermal vias. Input FET source faces battery.

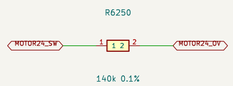


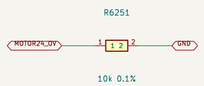


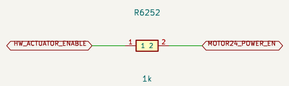


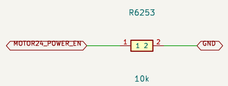




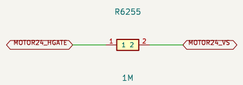




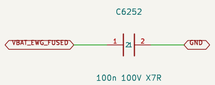






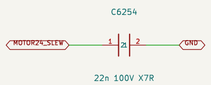


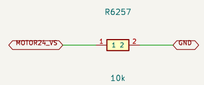












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UNQUALIFIED: no manufacturing or engine approval

Sheet: /motor4/motor24 power/
File: motor24_power.kicad_sch

Title: MOTOR24: independent protected feed

Size: A2 | Date:

KiCad E.D.A. 9.0.9

Rev: Engineering draft 0.1

Id: 49/49